

Load Balancing Algorithm Comparison Report

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Executive Summary

This report compares multiple load balancing algorithms. Key findings:

PPO Performance: +0.00% reward improvement

CPU Usage Reduction: +0.00%

Memory Usage Reduction: +0.00%

Bandwidth Usage Reduction: +0.00%

Performance Metrics Comparison

ward	std_reward	avg_cpu_usage	avg_mem_usage	avg_bw_usage	avg_score	vm_distribu
70	0.0747	0.7601	0.0000	0.0000	1.7936	{'laos-1': 8, 'laos-2': 8, 'laos
n	nan	nan	nan	nan	None	{'laos-4': 8
48	0.0176	0.6736	0.0000	0.0000	1.6810	{'laos-2': 30, 'laos

Dataset-level Metrics (selected window)

Metric	Value
makespan_seconds	8.0
energy_joules_estimate	30994.94057884076
avg_resource_utilization	0.34512047633992254
avg_response_time	0.3428571428571428
avg_execution_time	0.7999999999999999
avg_task_completion_time	1.1428571428571428
degree_of_imbalance	0.007988854550297872

Algorithm Details

Baseline Algorithm

Total Requests: 32
Average Reward: 0.5470
Average CPU Usage: 76.01%
Average Memory Usage: 0.00%
Average Bandwidth Usage: 0.00%
Load Balance Index: 0.0025

PPO Algorithm

Total Requests: 8
Average Reward: N/A
Average CPU Usage: N/A
Average Memory Usage: N/A
Average Bandwidth Usage: N/A
Load Balance Index: N/A

Chaos BWO Algorithm

Total Requests: 32
Average Reward: 0.6448
Average CPU Usage: 67.36%
Average Memory Usage: 0.00%
Average Bandwidth Usage: 0.00%
Load Balance Index: 0.0013